

Temporal specificity is not confirmatory yet

RULING - DEMOTE TO EXPLORATORY. The inherited centered object can support a scene-count upper bound, but it cannot verify a matched pre-pass/post-pass pair.

56	66	0	0
required city-year-window-sensor HLS rows	legacy centered axis records	verified pre/post lag records	stored optical acquisition IDs

What was run

Frozen rule V6.2-D009 was applied to the primary 15-degree historical pass denominator. The complete reporting grid covers Phoenix and Los Angeles, 2019-2025, both frozen season windows, and HL30.002 plus HLSS30.002. Only coordinates, attributes, and observed_valid_count were opened from the inherited centered object.

What came out

The object reports COPENICUS/S2_SR_HARMONIZED, Phoenix only, 2 June to 29 September 2023. It is therefore not the frozen HLS V2 product. Every one of its 66 axis records has a spatial median centered scene count of at least two, but no source date, sensor record, or optical acquisition identifier is stored.

Why count >=2 is not a pair

A centered count has no sign. Two contributing scenes may both be before the pass, both after it, or at very different lags. It also cannot show whether the same optical acquisition was reused across many thermal passes. Treating count >=2 as a verified pair would create timing evidence that is not present in the file.

Interpretation boundary

The count is feasibility-only. The centered object is not the final antecedent exposure and is not the final future placebo. Both must be rebuilt separately from raw HLS with one-sided timing. No optical index value, ECOSTRESS LST value, or new v6.2 coefficient was opened. Gate A remains unauthorised.

Legacy centered count inventory

Window	D1c candidates	Count >=2 upper bound	All legacy axes	Matched axes	Verified pairs
Primary	2	2	27	3	Not estimable
Sensitivity	3	3	66	5	Not estimable

D1c candidates are Phoenix 2023 passes in the inherited 15-degree metadata census. The all-axis column describes the broader legacy object; matched axes are records that join to those candidate pass orbits. Neither column contains optical source timing.

What would restore a confirmatory timing test

Frozen verified-pair rule: distinct acquisitions from the same sensor; one at least 24 hours before and one at least 24 hours after the thermal pass; each 1-15 days away; absolute lags within 3 days of one another. Choose the least-imbalanced, then shortest pair.

Candidate-pass denominator

Year	PHX primary	PHX sensitivity	LA primary	LA sensitivity
2019	2	5	4	2
2020	0	1	3	0
2021	0	3	5	2
2022	0	2	2	0
2023	2	3	5	2
2024	1	1	1	0
2025	0	2	5	2

Counts use the D1b historical quality-and-exact-weather-complete pass census at the primary 15-degree geometry set. They are feasibility denominators, not the new v6.2 sample.

Ruling

Demote the temporal-specificity diagnostic to exploratory. The result is about missing timing provenance, not proof that usable pre/post imagery is absent. A numeric matched-pair share is deliberately left blank rather than reported as zero.

What remains unresolved

1. Build a source-level HLSL30.002 and HLSS30.002 ledger for both cities and all years, retaining acquisition ID, source date, sensor, QA, and thermal-pass reuse.
2. Apply the frozen matched-lag rule and rerun the 56-row table plus the timing histogram.
3. Rebuild final antecedent and future W separately from raw HLS. Never promote the centered composite to either role.

Files to inspect

tables/d1c_leadlag.csv contains every required HLS stratum and two labeled legacy audit rows. figures/d1c_timing.png shows the explicit no-verifiable-lag state. data/d1c_centered_count_audit.json records every permitted count-only axis read.